



University of sigmas

Oiowe chady

Adam Biel, Mateusz Jurach, Karol Źeleźnik

2024-10-18

1 **Templatki**

2 **Wzorki**

3 **Matma**

4 **Struktury danych**

5 **Grafy**

6 **Tekstówki**

7 **Flow**

8 **Optymalizacje**

Templatki (1)

.bashrc 11 lines

```
c() {
    g++ -std=c++20 -Wall -Wno-sign-conversion -Wextra -Wshadow -
        Wconversion -Wfloat-equal \
        -fsanitize=address,undefined -DLOCAL -DDEBUG -D_GLIBCXX_DEBUG
        -ggdb3 $1.cpp -o $1
}

nc() {
    g++ -std=c++20 -DLOCAL -O3 -static $1.cpp -o $1
}
```

```
alias cp='cp -i'
alias mv='mv -i'
```

.vimrc 7 lines

```
set nocompatible
filetype indent on
set autoindent
set number
syntax on
set ts=4 sw=4
set rnu
```

main.cpp 24 lines

```
#include <bits/stdc++.h>
using namespace std;
#ifdef DEBUG
auto&operator<<((auto &o, pair<auto, auto> p) {o << "(" << p.
    first << ", " << p.second << " "}; return o;}
auto operator<<((auto&o,auto x)->decltype(x.end()),o){o<<"{"; for
    (auto e : x) o<<e<<" "; return o<<"}";}
#define debug(X) cerr << "["#X"]": " << X << '\n';
#else
#define cerr if(0)cout
#define debug(X) ;
#endif
using ll = long long;
#define all(v) (v).begin(), (v).end()
#define ssize(x) int(x.size())
#define fi first
#define se second
```

```
1 #define mp make_pair
1 #define eb emplace_back
1
1 int main() {
1     ios_base::sync_with_stdio(false); cin.tie(nullptr);
1
1     return 0;
1 }
3
```

```
6 gen.cpp
6 Opis: Dodatek do generowania testów, lub innego losowania. Ziarno można
6 zmieniać np. rng.seed(atoi(argv[1]));
7
7 mt19937 rng(random_device{}());
7 int rd(int l, int r) {
7     return uniform_int_distribution<int>(l, r)(rng);
7 }
```

7 **Wzorki (2)**

2.1 Fibonacci

$f_0 = 0, f_1 = 1, f_n = f_{n-1} + f_{n-2}$, dla $n \geq 2$.

$$f_n = \frac{\alpha^n - \beta^n}{\sqrt{5}} = \frac{\alpha^n - \beta^n}{\alpha - \beta},$$

dla $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}$. Jako, że $|\beta| < 1$ to

$$f_n \approx \frac{1}{\sqrt{5}} \alpha^n.$$

Niech $F(x) = \sum_{n=0}^{\infty} f_n x^n$, wtedy

$$F(x) = \frac{x}{1 - x - x^2}.$$

$$f_1 + f_2 + \dots + f_n = f_{n+2} - 1$$

$$\text{NWD}(f_n, f_m) = f_{\text{NWD}(n, m)}.$$

2.2 Liczby Catalana

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0$$

$$C_{n+1} = \frac{2(2n+1)}{n+2} C_n$$

2.3 Funkcje tworzące

$$\frac{1}{1 - ax} = \sum a^n x^n$$

$$F(x) = \frac{x}{1 - x - x^2}$$

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}$$

$$\exp(x) = \sum_{n \geq 0} \frac{x^n}{n!}$$

$$-\ln(1 - x) = \sum_{n \geq 1} \frac{x^n}{n}$$

$$\frac{1}{(1 - x)^k} = \sum_{n \geq 0} \binom{n + k - 1}{n} x^n$$

$$\prod_{n \geq 1} (1 - x^n) = 1 + \sum_{k \geq 1} (-1)^k \left(x^{\frac{k(3k+1)}{2}} + x^{\frac{k(3k-1)}{2}} \right)$$

2.4 Geometria

2.4.1 Obrót płaszczyzny wokół punktu (0,0).

$$R_\theta \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

W szczególności

$$R_{\frac{\pi}{4}} \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} x + y \\ x - y \end{bmatrix}.$$

Matma (3)

ext-euclid.cpp

Opis: $O(\log n)$. Zwraca trójkę [d, x, y], że $d = \text{NWD}(a, b)$ i $ax + by = d$.

```
tuple<ll, ll, ll> euclid(ll a, ll b) {
    if (b == 0) return make_tuple(a, 1, 0);
    auto [d, x, y] = euclid(b, a%b);
    return make_tuple(d, y, x - (a/b) * y);
}
```

crt.cpp

Opis: $O(\log n)$. Zwraca takie x , że $x \equiv a \pmod m$ i $x \equiv b \pmod n$. Mogą nie być względnie pierwsze ale wtedy assert wywali.

```
ll crt(ll a, ll m, ll b, ll n) {
    if (n > m) swap(a, b), swap(m, n);
    auto [d, x, y] = euclid(m, n);
    assert((a - b) % d == 0);
    ll ret = (b - a) % n * x % n / d * m + a;
    return ret < 0 ? ret + m * n / d : ret;
}
```

xor-base.cpp Opis: $O(nB)$ gdzie B to liczba bitów. Zwraca minimalny zbiór b taki że każdy element z x można zapisać jako xor pewnych elementów b. <pre>vector<ll> xor_base(const vector<ll> &x) { vector<ll> basis; for (ll a: x) { ll A = a; for (ll b: basis) A = min(A,A^b); if (A) basis.push_back(A); } return basis; }</pre> fft.cpp Opis: Mnożenie wielomianów w $O(n \log n)$. <pre>using ld = long double; using cd = complex<long double>; const double PI = acos(-1); void fft(vector<cd> &a, int n, bool invert = false) { assert((n&(n-1)) == 0); a.resize(n); for (int i = 1, j = 0; i < n; ++i) { int bit = n/2; for (; bit&j; bit /= 2) j ^= bit; j ^= bit; if (i < j) swap(a[i], a[j]); } for (int len = 2; len <= n; len *= 2) { ld ang = 2 * PI / len * (invert ? -1 : 1); cd r_len(cos(ang), sin(ang)); for (int i = 0; i < n; i += len) { cd r(1); for (int j = 0; j < len/2; ++j, r *= r_len) { cd u = a[i + j], v = r * a[i + j + len/2]; a[i+j] = u + v; a[i+j+len/2] = u - v; } } } if (invert) for (cd &x : a) x /= n; } vector<ld> conv(vector<ld> a, vector<ld> b) { if (a.empty() b.empty()) return {}; vector<cd> fa(all(a)), fb(all(b)); int n = ssize(a) + ssize(b) - 1; int sz = 1; while (sz < ssize(a) + ssize(b) - 1) sz *= 2; fft(fa, sz); fft(fb, sz); for (int i = 0; i < sz; ++i) fa[i] *= fb[i]; fft(fa, sz, true); vector<ld> result(n); for (int i = 0; i < n; ++i) result[i] = round(fa[i].real()); return result; }</pre>	mod.cpp Opis: operacje mod 998244353. <pre>const int mod = 998244353; int add(int a, int b) { a += b; return a >= mod ? a-mod : a; } int sub(int a, int b) { return add(a, mod-b); } int mul(int a, int b) { return int(((ll)a*b)%mod); } int fpow(int a, int b) { int res = 1; while (b) { if (b&1) { res = mul(a, res); b--; } a = mul(a, a); b /= 2; } return res; } int inv(int x) { return fpow(x, mod-2); }</pre> ntt.cpp Opis: $O(n \log n)$. Mnożenie wielomianów mod 998244353. <pre>#include "mod.cpp" //keep-include const int g = 3; const int g_1 = 332748118; using vi = vector<int>; void ntt(vi &a, int n, bool invert = false) { assert((n&(n-1)) == 0); a.resize(n); for (int i = 1, j = 0; i < n; ++i) { int bit = n/2; for (; bit&j; bit /= 2) j ^= bit; j ^= bit; if (i < j) swap(a[i], a[j]); } for (int len = 2; len <= n; len *= 2) { int r_len = fpow(invert ? g_1 : g, (mod-1) / len); for (int i = 0; i < n; i += len) { int r = 1; for (int j = 0; j < len/2; ++j, r = mul(r, r_len)) { int u = a[i + j], v = mul(r, a[i + j + len/2]); a[i+j] = add(u, v); a[i+j+len/2] = sub(u, v); } } } if (invert) {</pre>	<pre> int inv_n = inv(n); for (int &x : a) x = mul(x, inv_n); } } vi conv(vi a, vi b) { if (a.empty() b.empty()) return {}; int n = ssize(a) + ssize(b) - 1; int sz = 1; while (sz < ssize(a) + ssize(b) - 1) sz *= 2; ntt(a, sz); ntt(b, sz); for (int i = 0; i < sz; ++i) a[i] = mul(a[i], b[i]); ntt(a, sz, true); a.resize(n); return a; }</pre> poly.cpp Opis: Operacje na wielmianach mod 998244353. Można przepisać co się chce, przy funkcjach jest napisane co potrzeba. deriv, integr w $O(n)$. inv, log, exp w $O(n \log n)$. <pre>#include "ntt.cpp" //keep-include using vi = vector<int>; vi mod_xn(const vi &v, int n) { //ZAWSZE return vi(v.begin(), v.begin() + min(n, ssize(v))); } vi add(vi a, const vi &b) { a.resize(max(ssize(a), ssize(b))); for (int i = 0; i < ssize(b); ++i) a[i] = add(a[i], b[i]); return a; } vi sub(vi a, const vi &b) { //ZAWSZE a.resize(max(ssize(a), ssize(b))); for (int i = 0; i < ssize(b); ++i) a[i] = sub(a[i], b[i]); return a; } vi deriv(vi p) { for (int i = 1; i < ssize(p); ++i) p[i] = mul(p[i], i); if (ssize(p)) p.erase(p.begin()); return p; } vi integr(vi p) { for (int i = 0; i < ssize(p); ++i) p[i] = mul(p[i], inv(i+1)); p.insert(p.begin(), 0); return p; } vi inv(vi p, int n) { assert(ssize(p) && p[0] != 0); if (n == 1) return {inv(p[0])}; p.resize(n); vi _p = p; for (int i = 1; i < n; i += 2) _p[i] = sub(0, p[i]); vi t = mod_xn(conv(p, _p), n); for (int i = 1; i < n; i += 2) assert(t[i] == 0); for (int i = 0; i < n; i += 2) t[i/2] = t[i];</pre>
--	---	---

```
t = inv(t, (n+1)/2);
t.resize(n);
for (int i = (n-1)/2; i >= 0; --i) {
    t[2*i] = t[i];
    if(i) t[i] = 0;
}
return mod_xn(conv(_p, t), n);
}

vi log(const vi &p, int n) { //DERIV, INV, INTEGR
    assert(ssize(p) && p[0] == 1);
    return integr(mod_xn(conv(deriv(mod_xn(p, n)), inv(p, n)), n
        -1));
}

vi exp(const vi &p, int n) { //ADD, LOG
    assert(p.empty() || p[0] == 0);
    vi q = {1};
    for (int x = 1; x < n; x *= 2)
        q = mod_xn(conv(q, add(sub({1}, log(q, 2*x)), mod_xn(p, 2*x
            )), 2*x);
    q.resize(n);
    return q;
}

vi powi(vi p, int k, int n) { //LOG, EXP
    p = mod_xn(p, n);
    int cnt = 0;
    while(cnt < ssize(p) and !p[cnt])
        ++cnt;
    if(ll(cnt) * k >= n)
        return {};
    p.erase(p.begin(), p.begin() + cnt);
    if(p.empty())
        return k ? vi{} : vi{1};
    int powi0 = fpow(p[0], k);
    int inv0 = inv(p[0]);
    for (int &x : p) x = mul(x, inv0);
    p = log(p, n - cnt * k);
    for (int &x : p) x = mul(x, k);
    p = exp(p, n - cnt * k);
    for (int &x : p) x = mul(x, powi0);
    vi t(cnt * k, 0);
    p.insert(p.begin(), t.begin(), t.end());
    return p;
}

vi sqrt(const vi &p, int n) { //POW
    assert(ssize(p) && p[0] == 1);
    return powi(p, inv(2), n);
}
```

poly2d.cpp

Opis: Operacje na wielmianach dwóch zmiennych mod 998244353. Mnożenie w $O(mn \log(m+n))$. Dzielenie zwraca vector współczynników przy x^{n-1} , w czasie $O(mn \log^2(n+m))$.

```
#include "poly.cpp" //keep-include
```

```
void ntt(vector<vi> &a, int n, int m, bool inverse = false) {
    a.resize(n);
    for (int i = 0; i < ssize(a); ++i)
        ntt(a[i], m, inverse);
    for (int j = 0; j < ssize(a[0]); ++j) {
        vi c;
        for (int i = 0; i < ssize(a); ++i)
            c.eb(a[i][j]);
        ntt(c, n, inverse);
        for (int i = 0; i < ssize(a); ++i)
```

```
        a[i][j] = c[i];
    }
}

vector<vi> conv(vector<vi> a, vector<vi> b, int n, int m) {
    if (a.empty() || b.empty()) return {};
    int l1 = ssize(a) + ssize(b) - 1, sz1 = 1 << __lg(2*l1-1);
    int l2 = ssize(a[0]) + ssize(b[0]) - 1, sz2 = 1 << __lg(2*l2
        -1);

    ntt(a, sz1, sz2);
    ntt(b, sz1, sz2);
    for (int i = 0; i < sz1; ++i)
        for (int j = 0; j < sz2; ++j) a[i][j] = mul(a[i][j], b[i][j
            ]);
    ntt(a, sz1, sz2, true);
    a.resize(n);
    for (int i = 0; i < n; ++i) a[i].resize(m);
    return a;
}

vi div(vector<vi> b, vector<vi> a, int n, int m) {
    if (n == 1) {
        return conv(b[0], inv(a[0], m));
    }
    vector<vi> q = a;
    for (int i = 1; i < n; i += 2)
        for (int j = 0; j < ssize(a[i]); ++j) q[i][j] = sub(0, q[i
            ][j]);

    vector<vi> v = conv(a, q, n, 2*m);
    debug(v);
    q = conv(b, q, n, 2*m);
    for (int i = 0; i < ssize(v); i += 2) {
        for (int j = 0; j < ssize(v[i]); ++j) {
            v[i/2][j] = v[i][j];
        }
    }
    v.resize((n+1)/2);

    if (n%2 == 0) {
        for (int i = 1; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize(n/2);
        return div(q, v, n/2, 2*m);
    }
    else {
        for (int i = 0; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize((n+1)/2);
        return div(q, v, (n+1)/2, 2*m);
    }
}
```

lis.cpp

Opis: $O(n \log n)$. Zwraca długość najdłuższego podciągu rosnącego.

```
int LIS(vector<int> A)
{
    vector<int> vec;
    for(auto& v : A)
        if(vec.size() == 0 || v > vec.back())
            vec.push_back(v);
        else
            *lower_bound(vec.begin(), vec.end(), v) = v;
    return vec.size();
}
```

mo.cpp

Opis: $O(n \log n)$. Sort do MO. Sortuje przedziały $[a.x, a.y]$.

```
const int k = 500;
sort(events.begin(), events.end(), [&](type a, type b){
    if(a.x/k < b.x/k) return true;
    if(a.x/k > b.x/k) return false;
    if((a.x/k)%2 == 1)
        return a.y < b.y;
    else
        return a.y > b.y;
});
```

sos.cpp

Opis: $O(l \cdot 2^l)$. Zwraca $sos[mask]$ równe sumie $s[m]$ po podzbiorach $mask$.

```
vector<int> SOS(int l, vector<int>& s)
{
    vector<int> sos(1<<l);
    for(int mask = 0; mask<(1<<l); mask++)
        sos[mask] = s[mask];
    for(int i=0; i<l; i++)
        for(int mask=0; mask<(1<<l); mask++)
            if((mask & (1<<i)) != 0)
                sos[mask] += sos[mask^(1<<i)];

    return sos;
}
```

Struktury danych (4)

treap.cpp

Opis: $O(\log n)$ czasowo. Treap z operacjami na (multi)secie. $kth(t, k)$ zwraca k -tą najmniejszą liczbę.

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int val, rank;
        int size;
        Node(): left(nullptr), right(nullptr), val(0), rank(int(rng
            )), size(0) {}
    };
    using pNode = Node*;

    Node pool[1'000'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }
```

```
pNode merge(pNode a, pNode b) {
    if(a == nullptr) return b;
    if(b == nullptr) return a;
```

```
    if(a->rank < b->rank) {
        a->right = merge(a->right, b);
        upd(a);
        return a;
    }
    else {
        b->left = merge(a, b->left);
```

```
        upd(b);
        return b;
    }
}

pair<pNode, pNode> split(pNode t, int x) {
    if(t == nullptr) return {nullptr, nullptr};

    if(t->val <= x) {
        auto [a, b] = split(t->right, x);
        t->right = a;
        upd(t);
        return {t, b};
    }
    else {
        auto [a, b] = split(t->left, x);
        t->left = b;
        upd(t);
        return {a, t};
    }
}

void insert(pNode &root, int x) {
    pNode u = pool + (curr++);
    u->val = x;
    u->size = 1;
    auto [a, b] = split(root, x);
    root = merge(merge(a, u), b);
}

bool find(pNode t, int x) {
    if(t == nullptr) return false;
    if(x == t->val) return true;
    if(x < t->val) return find(t->left, x);
    else return find(t->right, x);
}

void erase(pNode root, int x) {
    auto [ab, c] = split(root, x);
    auto [a, b] = split(ab, x-1);
    root = merge(a, c);
}

int kth(pNode t, int k) {
    assert(k >= 0);
    if(sz(t->left) == k-1) return t->val;
    if(k > sz(t->left)) return kth(t->right, k-1-sz(t->left));
    else return kth(t->left, k);
}
}
```

implicit-treap.cpp

Opis: $O(\log n)$ czasowo. Implicit treap z podstawowymi operacjami, przykładowo odwracanie na przedziale (może być dowolna operacja). insert(root, k, x) wstawia element o wartości x w k -te miejsce tablicy. rev(root, l, r) odwraca przedział $[l;r]$ (indeksy od 0).

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int rank;
        int size;
        bool rev;
        int val;
        Node(): left(nullptr), right(nullptr), rank(int(rng())),
            size(0), rev(false), val(0) {}
    };
    using pNode = Node*;
```

```
Node pool[300'003];
int curr = 0;

int sz(pNode t) {
    return t == nullptr ? 0 : t->size;
}

void upd(pNode t) {
    t->size = sz(t->left) + sz(t->right) + 1;
}

void push_lazy(pNode t) {
    if(t->rev == false) return;
    if(t->left) t->left->rev = !t->left->rev;
    if(t->right) t->right->rev = !t->right->rev;
    swap(t->left, t->right);
    t->rev = false;
}

pNode merge(pNode a, pNode b) {
    if(a == nullptr) return b;
    if(b == nullptr) return a;

    push_lazy(a);
    push_lazy(b);
    if(a->rank < b->rank) {
        a->right = merge(a->right, b);
        upd(a);
        return a;
    }
    else {
        b->left = merge(a, b->left);
        upd(b);
        return b;
    }
}

pair<pNode, pNode> split(pNode t, int x) {
    if(t == nullptr) return {nullptr, nullptr};

    push_lazy(t);
    if(sz(t->left)+1 <= x) {
        auto [a, b] = split(t->right, x-(sz(t->left)+1));
        t->right = a;
        upd(t);
        return {t, b};
    }
    else {
        auto [a, b] = split(t->left, x);
        t->left = b;
        upd(t);
        return {a, t};
    }
}

pNode build(vector<int> a) {
    pNode root = nullptr;
    int n = ssize(a);
    for(int i = 0; i < n; ++i) {
        pNode u = pool + i;
        u->size = 1;
        u->val = a[i];
        root = merge(root, u);
    }
    curr += n;
    return root;
}
```

```
void insert(pNode &root, int k, int x) {
    auto [a, b] = split(root, k-1);
    Node *u = pool + (curr++);
    u->size = 1;
    u->val = x;
    root = merge(merge(a, u), b);
}

void reverse(pNode &root, int l, int r) {
    auto [ab, c] = split(root, r+1);
    auto [a, b] = split(ab, l);
    b->rev = !b->rev;
    root = merge(a, merge(b, c));
}

void print(pNode t) {
    if(t == nullptr) return;
    push_lazy(t);
    print(t->left);
    cout << t->val << ' ';
    print(t->right);
}
}
```

uf-rollback.cpp

Opis: $O(\log n)$ czasowo. Union find z roll backami – roll_back(x) cofa ostatnie x operacji unionn (union wierzchołków z tego samego zbioru się nie liczy).

```
struct dynamicUF {
    vector<int> e;
    dynamicUF(int n):e(n, -1) {}
    stack<array<int, 3>> s;

    int get(int a) {
        while(e[a] >= 0) a = e[a];
        return a;
    }

    int unionn(int a, int b) {
        a = get(a); b = get(b);
        if(a == b) return 0;
        if(-e[a] > -e[b]) swap(a, b);
        s.push({a, b, e[a]});
        e[b] += e[a];
        e[a] = b;
        return 1;
    }

    void roll_back(int m = 1) {
        while (m--) {
            auto [a, b, sz] = s.top();
            s.pop();
            e[b] -= sz;
            e[a] = sz;
        }
    }
};
```

fenwick-tree.cpp

Opis: (Binary Indexed Tree) $O(\log N)$, indeksowane od 0. Update w punkcie, zapytanie na przedziale

```
struct BIT {
    vector<int> tree;
    int N;
    BIT() {N = 0;}
    BIT(int n) {
        N = n;
```

```
        tree.resize(N);
    }
    void update(int p, int x) {
        for (++p; p <= N; p += (p&p)) tree[p-1] += x;
    }
    int sum(int r) {
        int res = 0;
        for (; r > 0; r -= (r&r)) res += tree[r-1];
        return res;
    }
    int query(int l, int r) {return sum(r+1)-sum(l);}
    int lower_bound(int sum) {
        int pos = 0;
        for (int pw = 1<<25; pw; pw>>=1) {
            int npos = pos+pw;
            if (npos <= N && tree[npos-1] < sum) {
                pos = npos; sum -= tree[pos-1];
            }
        }
        return pos;
    }
};
```

range.cpp

Opis: $O(\log n)$. Dodaj na przedziale, podaj wartość w punkcie.

```
#include "fenwick-tree.cpp" //keep-include

struct RangeAdd {
    BIT diff;
    RangeAdd(int n): diff(n) {}
    void update(int l, int r, int x) {
        if (l) diff.update(l-1, x);
        diff.update(r, -x);
    }

    int query(int p) {
        return diff.sum(p);
    }
};
```

fenwick-tree2d.cpp

Opis: $O(\log^2 n)$. Dodaj w punkcie, suma na prostokącie.

```
struct BIT2d {
    vector<vector<int>>> tree;
    int n, m;
    BIT2d() {n = m = 0;}
    BIT2d(int N, int M) {
        n = N; m = M;
        tree.resize(n, vector<int>(m));
    }
    void update(int x, int y, int v) {
        for (++x; x <= n; x += (x&-x))
            for (int p = y+1; p <= m; p += (p&p))
                tree[x-1][p-1] += v;
    }
    int sum(int x, int y) {
        int res = 0;
        for (; x; x -= (x&-x))
            for (int j = y; j; j -= (j&-j))
                res += tree[x-1][j-1];
        return res;
    }
    int query(int x1, int y1, int x2, int y2) {return sum(x2+1,
        y2+1)-sum(x1, y2+1)-sum(x2+1, y1)+sum(x1, y1);}
};
```

pbds.cpp

Opis: Ordered set.

```
#include <ext/pb_ds/assoc_container.hpp> //keep-include
#include <ext/pb_ds/tree_policy.hpp> //keep-include
using namespace __gnu_pbds;

template<typename T> using ordered_set = tree<
T,
null_type,
less<T>,
rb_tree_tag,
tree_order_statistics_node_update>;

void test() {
    ordered_set<int> S;
    S.insert(1);
    cout << *S.find_by_order(0) << '\n'; //zwraca k-ty element (
indeksowane od 0)
    cout << (S.end() == find_by_order(1)) << '\n';
    cout << S.order_of_key(4) << '\n'; //zwraca liczbe elementow
mniejszych od x
}
```

merge-sort-tree.cpp

Opis: Preprocessing $O(n)$, query w $O(\log n \cdot T(n))$, gdzie $T(n)$ to czas operacji na przedziale bazowym. Pamięć $O(n \log n)$.

```
const int base = 1<<20;
struct mergeSortTree
{
    vector<vector<int>>> tree;
    void build(int w, vector<int>& A)
    {
        if(w >= base)
        {
            if(w - base < A.size()) tree[w].push_back({A[w-base]});
            return;
        }
        build(w*2, A);
        build(w*2+1, A);
        merge(tree[w*2].begin(), tree[w*2].end(),
            tree[w*2+1].begin(), tree[w*2+1].end(),
            back_inserter(tree[w]));
        return;
    }
    mergeSortTree(vector<int>& A)
    {
        tree.resize(base*2);
        build(1, A);
    }
    int query(int w, int p, int k, int x, int y, int val)
    {
        if(p > y || k < x) return 0;//poza zapytaniem
        if(x <= p && k <= y)
        {
            auto it = upper_bound(tree[w].begin(), tree[w].end(), val
                );
            //tutaj robimy cos na przedziale bazowym zapytania
            return distance(tree[w].begin(), it);//returnujemy wynik
tego przedzialu
        }
        return query(w*2, p, (p+k)/2, x, y, val) +
            query(w*2+1, (p+k)/2+1, k, x, y, val); //tu dowolna
operacja (max, +, -, *, czy cokolwiek)
    }
    int query(int x, int y, int val) {return query(1, 0, base-1,
        x, y, val);}//ile mniejszych od val na (x, y)
};
```

wavelet.cpp

Opis: $O(\log^2 \text{ALPH})$ czasowo, $O(n \log \text{ALPH})$ pamięciowo. Constructor przyjmuje ciąg na którym rozpinamy drzewo, rank(l, r, k) zwraca k -tą najmniejszą liczbę na przedziale $[l; r]$. (Uwaga!) pamięć około 262 Mb, dla $N, Q \leq 2 \cdot 10^5$.

```
//funkcja cnt jest w O(\log) co daje cale operacje w O(\log^2),
da sie cnt w czasie stalym — do dopisania
struct wavelet_tree {
    const static int ALPH = (1<<18)-1;
    struct Node {
        int low, high, mid;
        vector<int> v;
        Node *left, *right;

        Node() {
            low = 0, high = ALPH;
            left = right = nullptr;
            mid = (low+high)/2;
        }

        Node(const vector<int> &x, const vector<int> &t, int a =
            0, int b = ALPH): low(a), high(b), v(x) {
            mid = ((1l)low+high)/2;
            if(ssize(x) && low != high) {
                vector<int> div[2];
                for(auto e : v) div[t[e]<=mid].eb(e);
                if(ssize(div[1])) left = new Node(div[1], t, low, mid);
                else left = nullptr;
                if(ssize(div[0])) right = new Node(div[0], t, mid+1,
                    high);
                else right = nullptr;
            }
            else {
                left = right = nullptr;
            }
        }
    };

    Node *root;
    wavelet_tree(const vector<int> &b) {
        vector<int> idx;
        REP(i, ssize(b)) idx.eb(i);
        root = new Node(idx, b);
    };

    int cnt(Node *v, int l, int r) {
        if(v == nullptr) return 0;
        return int(upper_bound(all(v->v), r)-lower_bound(all(v->v),
            l));
    }

    int rank(int l, int r, int k, Node *v) {
        if(v->low == v->high) return v->low;
        if(cnt(v->left, l, r) >= k) return rank(l, r, k, v->left);
        else return rank(l, r, k-cnt(v->left, l, r), v->right);
    }
    int rank(int l, int r, int k) {
        return rank(l, r, k, root);
    }

    void clear(Node *v) {
        if(v->left != nullptr) clear(v->left);
        if(v->right != nullptr) clear(v->right);
        delete v;
    }
    ~wavelet_tree() {
        clear(root);
    }
};
```

```
};
```

Grafy (5)

5.1 Drzewo DFS

Mamy drzewo dfs z obliczonymi czasami wejścia w tablicy **pre** i funkcją

$$\text{low}[v] = \min \begin{cases} \text{pre}[v], \\ \text{pre}[u] & \text{dla krawędzi niedrzewowych } (u, v), \\ \text{low}[u] & \text{dla krawędzi drzewowych } (u, v). \end{cases}$$

Wierzchołek v jest punktem artykulacji jeżeli

$$\text{low}[u] \geq \text{pre}[v]$$

dla dowolnego u będącego synem v w drzewie dfs.

Krawędź (v, u) (v jest ojcem u) jest mostem, jeżeli jest krawędzią drzewową i

$$\text{low}[u] > \text{pre}[v].$$

hld.cpp

Opis: $O(n)$ preprocessing, update i query w $O(\log^2 n)$. Update to zmiana wartosci w wierzcholku, query to max na sciezce. Drzewo przedzialowe do dopisania tak jak w komentarzu.

```
const int inf = 1e9+10;

//struct segtree {
//    segtree(int n = 0) {}
//    void update(int v, int x) {}
//    int query(int l, int r) {}
//};

struct HLD {
    int n;
    vector<vector<int>> g;
    segtree t;
    int timer;

    vector<int> sz, d, parent, path, c, pre;

    HLD(const vector<vector<int>> &graph): n(ssize(graph)), g(
        graph) {
        sz = d = parent = path = c = pre = vector<int> (n);

        d[0] = 0;
        dfs(0);

        path[0] = 0;
        timer = 0;
        make_hld(0);
        t = segtree(timer);
    }

    void dfs(int v, int p = -1) {
        parent[v] = p;
        sz[v] = 1;
        c[v] = -1;
        for (auto u : g[v]) {
```

```
            if (u == p) continue;
            d[u] = d[v] + 1;
            dfs(u, v);
            sz[v] += sz[u];
            if (c[v] == -1 || sz[c[v]] < sz[u]) c[v] = u;
        }
    }

    void make_hld(int v, int p = -1) {
        pre[v] = timer++;
        if (c[v] != -1) {
            path[c[v]] = path[v];
            make_hld(c[v], v);
        }
        for (auto u : g[v]) {
            if (u == p || u == c[v]) continue;
            path[u] = u;
            make_hld(u, v);
        }
    }

    void update(int v, int x) {
        t.update(pre[v], x);
    }

    int query(int a, int b) {
        int res = -inf;
        while (path[a] != path[b]) {
            if (d[path[a]] < d[path[b]]) swap(a, b);
            res = max(res, t.query(pre[path[a]], pre[a]));
            a = parent[path[a]];
        }
        if (d[a] < d[b]) swap(a, b);
        res = max(res, t.query(pre[b], pre[a]));
        return res;
    }
};
```

scc.cpp

Opis: silnie spójne składowe. Argument do constructora to graf skierowany w postaci `vector<vector<int>> v`. Oblicza `vector` indeksów spójnie składowej do której należy v o nazwie `scc`. Czas, pamiec $O(n + m)$.

```
struct SCC{
    vector<vector<int>>v,x;
    int n;
    queue<int>q;
    vector<bool>vis;
    int kt=0;
    vector<int>scc;
    SCC(const vector<vector<int>>&vv):v(vv),n(ssize(vv)){
        vis.resize(n,0);
        scc.resize(n);
        for(int i=0;i<n;i++){
            if(!vis[i])dfs0(i);
        }
        make_x();
        fill(all(vis),0);
        while(!q.empty()){
            if(!vis[q.front()]){
                dfs1(q.front());
                kt++;
            }
            q.pop();
        }
    }

    void dfs0(int a){
        vis[a]=1;
        for(auto b:v[a]){
```

```
            if(!vis[b])dfs0(b);
        }
        q.push(a);
    }
}

void make_x(){
    x.resize(n);
    for(int i=0;i<ssize(v);i++){
        for(auto b:v[i]){
            x[b].push_back(i);
        }
    }
}

void dfs1(int a){
    vis[a]=1;
    scc[a]=kt;
    for(auto b:v[a]){
        if(!vis[b])dfs1(b);
    }
}

};
```

bellman-ford.cpp

Opis: W $O(nm)$ oblicza tablice **D** z odległościami od s w grafie skierowanym ważonym (dopuszczane ujemne wagi). Po obliczeniu odległości `isNegCyc()` zwraca czy w grafie jest ujemny cykl.

```
const int INF = numeric_limits<int>::max();
struct edge
{
    int a, b, cost;
    edge(int _a, int _b, int _cost){a = _a; b = _b; cost = _cost;
}

};
struct Bellman_Ford
{
    int n;
    vector<edge> edges;
    vector<int> D;
    bool makeStep()
    {
        bool e = 0;
        for(auto v : edges)
            if(D[v.a] != INF)
                if(D[v.b] > D[v.a] + v.cost)
                    D[v.b] = D[v.a] + v.cost, e = true;
        return e;
    }
    Bellman_Ford(int _n, vector<edge>& _edges, int s)
    {
        n = _n; edges = _edges;
        D.resize(n+1, INF);
        D[s] = 0;
        while(_n-->1) makeStep();
    }
    bool isNegCyc()
    {
        return makeStep();
    }
}
};
```

5.2 Floyd-Warshall

$O(n^3)$. Oblicza odległość między każdą parą wierzchołków w skierowanym grafie ważonym (dopuszczalne wagi ujemne).

```
Dla k = 0 do n - 1
    Dla i = 0 do n - 1
        Dla j = 0 do n - 1
            Odleglosc[i, j] = min(Odleglosc[i, j],
Odleglosc[i, k] + Odleglosc[k, j])
```

Tekstówki (6)

z.cpp
Opis: $O(n)$. Zwraca tablice z, gdzie $z[i]$ jest równe najdluższemu wspól-nemu prefiksowi słowa s i sufiksu $[i, \dots, n - 1]$.

```
vector<int> z(string s) {
    int n = ssize(s);
    vector<int> z(n);

    int l = 0, r = 0;
    for (int i = 1; i < n; ++i) {
        if (i < r) z[i] = min(z[i-1], r-i);
        while (i + z[i] < n && s[i+z[i]] == s[z[i]]) ++z[i];
        if (i + z[i] > r) {
            r = i + z[i];
            l = i;
        }
    }
    return z;
}
```

suffix-array.cpp

Opis: (źródło: <https://github.com/tonowak/acmlib/blob/master/code/string/array-short/main.cpp>) $O(n \log n)$, zawiera posortowane suffixy, zawiera pusty suffix, $lcp[i]$ to lcp suffixu $sa[i - 1]$ i $sa[i]$, Dla $s = aabaaab$, $sa = \{7, 3, 4, 0, 5, 1, 6, 2\}$, $lcp = \{0, 0, 2, 3, 1, 2, 0, 1\}$

```
pair<vector<int>, vector<int>> suffix_array(vector<int> s, int
    alpha = 26) {
    ++alpha;
    for(int &c : s) ++c;
    s.emplace_back(0);
    int n = ssize(s), k = 0, a, b;
    vector<int> x(s.begin(), s.end());
    vector<int> y(n), ws(max(n, alpha)), rank(n);
    vector<int> sa = y, lcp = y;
    iota(sa.begin(), sa.end(), 0);
    for(int j = 0, p = 0; p < n; j = max(1, j * 2), alpha = p) {
        p = j;
        iota(y.begin(), y.end(), n - j);
        for (int i = 0; i < n; ++i) if(sa[i] >= j)
            y[p++] = sa[i] - j;
        fill(ws.begin(), ws.end(), 0);
        for (int i = 0; i < n; ++i) ws[x[i]]++;
        for (int i = 1; i <= alpha - 1; ++i) ws[i] += ws[i - 1];
        for(int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
        swap(x, y);
        p = 1, x[sa[0]] = 0;
        for (int i = 1; i <= n-1; ++i) a = sa[i - 1], b = sa[i], x[
            b] =
            (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
    }
    for(int i = 1; i <= n-1; ++i) rank[sa[i]] = i;
    for(int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
        for(k && k--, j = sa[rank[i] - 1];
            s[i + k] == s[j + k]; k++);
    lcp.erase(lcp.begin());
    return {sa, lcp};
}
```

Flow (7)

dinic.cpp

Opis: $O(V^2E)$, chyba że mamy sieć jednostkową (max flow na krawędzi to 1) to $O(E\sqrt{V})$.

```
const int INF = numeric_limits<int>::max();
struct flowEdge
{
    int u, v, cap, flow;
    flowEdge(int _u, int _v, int _cap, int _flow){u = _u; v = _v;
        cap = _cap; flow = _flow;}
    flowEdge() {}
};
struct dinic
{
    vector<flowEdge> edges;
    vector<vector<int> > idx;
    queue<int> q;
    vector<int> lvl;
    vector<int> ptr;
    int n, m;
    dinic(int _n, int _m, vector<pair<pair<int, int>, int> >&
        _edges) //krawedzie musza byc znormalizowane do DG
    {
        n = _n; m = _m;
        idx.resize(n+1);
        lvl.resize(n+1);
        ptr.resize(n+1);
        for(auto v : _edges)
        {
            edges.push_back(flowEdge(v.first.first, v.first.second, v
                .second, 0));
            edges.push_back(flowEdge(v.first.second, v.first.first,
                0, 0));
            idx[v.first.first].push_back(edges.size()-2);
            idx[v.first.second].push_back(edges.size()-1);
        }
    }
    bool bfs()
    {
        fill(lvl.begin(), lvl.end(), -1);
        q.push(0);
        lvl[0] = 0;
        while(!q.empty())
        {
            int a = q.front();
            q.pop();
            for(auto v : idx[a])
            {
                if(lvl[edges[v].v] != -1) continue;
                if(edges[v].cap - edges[v].flow < 1) continue;
                lvl[edges[v].v] = lvl[a]+1;
                q.push(edges[v].v);
            }
        }
        return lvl[n] != -1;
    }
    int dfs(int v, int aktmin)
    {
        if(v == n) return aktmin;
        if(aktmin == 0) return aktmin;
        for(int& i = ptr[v]; i<idx[v].size(); i++)
        {
            int id = idx[v][i];
            int nv = edges[id].v;
            if(lvl[v] + 1 != lvl[nv] || edges[id].cap - edges[id].
                flow < 1) continue;
```

```
        long long bFlow = dfs(nv, min(aktmin, edges[id].cap -
            edges[id].flow));
        if(bFlow == 0) continue;
        edges[id].flow += bFlow;
        edges[id^1].flow -= bFlow;
        return bFlow;
    }
    return 0;
}
int maxFlow()
{
    int res = 0;
    while(true)
    {
        if(!bfs()) break;
        fill(ptr.begin(), ptr.end(), 0);
        while(int flow = dfs(0, INF)) res += flow;
    }
    return res;
}
};
```

Optymalizacje (8)

fastio.cpp

Opis: fastio.

```
bool is_digit(char c) {
    return c >= '0';
}

int fastin(){
    int x = 0;
    int s = 1;
    char c;
    c = getchar();
    if (c == '-') s = -1;
    else x = c - '0';
    while (is_digit(c = getchar())) {
        x = x * 10 + c - '0';
    }
    return s*x;
}

void write_int(unsigned long x) {
    char t[19];
    int i = 0;
    do {
        int d = x%10;
        t[i++] = '0'+d;
        x /= 10;
    } while (x > 0) ;

    while (--i >= 0) {
        putchar_unlocked(t[i]);
    }
    putchar_unlocked('\n');
}
```